

# Haogao GU

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## EDUCATION

- **PhD (Infectious Disease), The University of Hong Kong** Sep. 2016 – Oct. 2020
- **BMed (Preventive Medicine), Sun Yat-sen University** Sep. 2011 – Jun. 2016

## EXPERIENCE

- **Associate Research Scientist (Mailman School of Public Health), Columbia University** Jun. 2026 – Present  
*PI: Sen Pei, Department of Environmental Health Sciences*
- **Research Assistant Professor (School of Public Health), The University of Hong Kong** Jul. 2024 – Jun. 2026
- **Visiting Scholar (Evolutionary dynamics of infectious diseases), Institut Pasteur** Dec. 2024 – Feb. 2025  
*Host: Sebastian Duchene, Department of Computational Biology*
- **Postdoctoral Fellow (School of Public Health), The University of Hong Kong** Jan. 2021 – Jun. 2024  
*Mentor: Leo Poon, Department of Laboratory Sciences*

## HONORS AND AWARDS

- **Gold Medal, 51st International Exhibition of Inventions at Geneva** Mar. 2026  
*Core member. Project: A broad-spectrum live-attenuated mucosal vaccine against beta-coronaviruses*
- **The HKU-Pasteur Fellowship (4th edition), HKU-Pasteur Research Pole** Apr. 2024
- **Faculty Outstanding Research Output Award, The University of Hong Kong** Feb. 2024  
*awarded twice, for two separate projects*
- **Outstanding Project Team, on COVID-19 research award, Food and Health Bureau (Hong Kong)** Nov. 2021
- **Postgraduate Scholarship, The University of Hong Kong** 2016–2020
- **Best Student Poster, International Symposium for One Health Research (Guangzhou)** 2014
- **Individual Scholarship, Sun Yat-sen University** 2013
- **The Third Prize Scholarship, Sun Yat-sen University** 2012
- **First Prize, Chinese National Olympiad in Biology** Sep. 2010  
*qualified for guaranteed admission to Sun Yat-sen University*

## GRANTS

- **General Research Fund (GRF), Research Grants Council, Hong Kong SAR** 2026–2027  
*Early Prediction of SARS-CoV-2 Variant Dynamics: Modelling Virus Interactions with Antibodies and ACE2 Receptor.*  
Role: **Co-I** (PI: Hsiang-Yu Yuan, City University of Hong Kong) \$840,217 (HKD)
- **RGC Postdoctoral Fellowship Scheme, University Grants Committee, Hong Kong SAR** 2023–2026  
*Quantifying attenuation effects caused by dinucleotide biases and codon pair biases for development of safe and scalable live attenuated virus vaccines.*  
Role: **PI** \$1,258,250 (HKD)
- **CEIRR Option (University of Georgia), NIH, United States** 2023–2025  
*Investigation of repeated influenza vaccination for epitope focusing.*  
Role: **Key personnel** (PI: Sophie Valkenburg) \$454,428 (USD)

## TEACHING

- **Postgraduate Certificate in Bioinformatics for MLT (LMLS6010), HKU SPACE** Jan. 2026  
*Lecturer for 12 classes on Programming for Biological Data*
- **Scientific Skills (LMLS6003), MSc in Biomedical Science, HKU SPACE** Oct. 2024  
*Lecturer for 6 classes on Data Handling in Bioinformatics*
- **Applied Medical Microbiology (LMLS6004), MSc in Biomedical Science, HKU SPACE** Nov. 2024  
*Lecturer for 2 classes on Bioinformatic Tutorial*

## PUBLICATIONS

### Summary

Papers in peer-reviewed journals: **46** (citations: **2559**; h-index: **20**)  
- first-/last- authored papers **16**

**Journal papers (in reverse chronological order):**

1. **Gu, H.**, Li, J., Sun, W., Li, M., Leung, K., Wu, J. T., Yuan, H.-Y., Wang, M. H., Yang, B., McKay, M. R., Ning, N. 2026 & Poon, L. L. M. *Optimizing global genomic surveillance for early detection of emerging SARS-CoV-2 variants*. **Nature Communications**. (2026)
2. Song, Z. Y., Zhang, J., Shum, M. H. H., **Gu, H.**, Shi, W., Siu, G. K. H., Poon, L. L. M., Wu, J. T., Leung, K. S. M. & Lam, T. T. Y. *Geographical dispersion and transmission of multiple SARS-CoV-2 variants in Hong Kong*. **Urban Informatics**, 5(1), 20. (2026)
3. **Gu, H.**, Li, M., Wang, R., Lee, L. K., Tam, W. Y., Ho, A. Y. M., Yau, M. C. Y., Yip, K. T., Wong, R. C. W., Wong, B. K. C., Luk, K. S., Lam, J. Y. W., Que, T. L., Chow, V. C. Y., Chau, S. K. Y., Duchene, S., Siu, G. K. H. & Poon, L. L. M. *Genomic Epidemiology of Hospital-Associated SARS-CoV-2 Clusters in Hong Kong During a Period of Relaxed Visitation (May–August 2022)*. **Influenza and Other Respiratory Viruses**, 20(3), e70249. (2026)
4. Ye, Y., Shuai, H., Song, Y., Shum, M. H.-H., Smith, D. K., **Gu, H.**, Zhu, H., Wu, J. T., Lewis, N. S., Bonfante, F., Holmes, E. C., Peiris, M., Poon, L. L.-M., Guan, Y. & Lam, T. T.-Y. *Genomic features associated with sustained mammalian transmission of avian influenza A viruses*. **Nature Microbiology**, s41564-025-02257-4. (2026)
5. Te, N., Chin, A. W. H., **Gu, H.**, Jia, J. Z., Huang, Y., Cheng, S. M. S., Tan, C. W., Zhao, J., Hui, K. P. Y., Chan, M. C. W., Feng, B., Quadeer, A. A., McKay, M. R., Wang, L., Valkenburg, S. A., Peiris, M. & Poon, L. L. M. *Broad beta-CoV immunity and transmission blockade by a single-dose live-attenuated vaccine with atypical codon usage*. **Proceedings of the National Academy of Sciences**, 123(4), e2518645123. (2026)
6. Tuohetesen, S., Qu, Y., Hopke, P. K., Zhang, K., Liu, Y., Lin, S., **Gu, H.**, Wang, X., Lau, S. S. S., Lin, X., Gao, X., Wu, Y., 2025 Zhou, X., Lin, Z., Zhang, M., Sun, Y., Liu, X., Chen, J. & Zhang, W. *Potential impact of maternal nighttime light exposure and its interaction with sociodemographic characteristics on the risk of various congenital heart diseases*. **Epidemiology**, 36(5), 625–635. (2025)
7. Zhang, Y., Wei, J., Chen, S., Benmarhnia, T., Zhang, K., Wang, X., Deng, X., **Gu, H.**, Lin, Z., Qu, Y., Xiao, J., Jiang, J., Du, Z., Zhang, W. & Hao, Y. *Individual and mixed associations between fine particulate matter components and hospital admissions for hypertension: Insights from a large-scale South Chinese cohort study*. **Sustainable Cities and Society**, 106293. (2025)
8. Wu, G., Wang, S., Wu, W., Benmarhnia, T., Lin, S., Zhang, K., Xue Romeiko, X., **Gu, H.**, Qu, Y., Xiao, J., Deng, X., Lin, Z., Du, Z., Zhang, W. & Hao, Y. *Potential causal links of long-term PM2.5 components exposure with diabetes incidence and mortality in the United States*. **Sustainable Cities and Society**, 106071. (2025)
9. Sun, W., Choy, K. T., Cheng, K. M., Brackman, C. J., Cheng, S. M. S., Sit, T. H. C., Anne, C. N., Sims, L. D., **Gu, H.**, Tang, A. W. Y., Wong, A. N. C., Tsang, A. T. L., Koo, J. C. T., Luk, L. L. H., Yen, H.-L., Peris, M. & Poon, L. L. M. *Detection and characterisation of high pathogenicity avian influenza virus (H5N1/H5N8) clade 2.3. 4.4 b, Hong Kong SAR, China, 2021 to 2024*. **Eurosurveillance**, 2400839. (2025)
10. Cao, Y., Chin, A. W. H., **Gu, H.**, Li, M., Gu, Y., Lau, S. P. N., Hui, K. P. Y., Chan, M. C. W. & Poon, L. L. M. *An interferon-stimulated long non-coding RNA USP30-AS1 as an immune modulator in influenza A virus infection*. **PLOS pathogens**, e1012854. (2025)
11. Wang, X., Zhang, Y., Benmarhnia, T., Zhang, K., Lin, S., Deng, X., **Gu, H.**, Wu, W., Lin, X., Du, Z., Wang, Y., Qu, Y., 2024 Lin, Z., Zhang, M., Sun, Y., Dong, G.-H., Wei, Y., Zhang, W. & Hao, Y. *The effect of residential greenness on cardiovascular mortality from a large cohort in South China: An in-depth analysis of effect modification by multiple demographic and lifestyle characteristics*. **Environment International**, 108894. (2024)
12. Nelson, C., Poon, L. L. M. & **Gu, H.** *Reply to: Population genetic considerations regarding the interpretation of within-patient SARS-CoV-2 polymorphism data*. **Nature Communications**, 15, 3239. (2024)
13. Jia, J.\*, Cohen, C.\*, **Gu, H.\***, McLean, M., Varadarajan, R., Bhandari, N., Peiris, M., Leung, G., Poon, L., Chung, A., Cowling, B., Leung, N., Tsang, T. & Valkenburg, S. *Influenza antibody breadth and effector functions are immune correlates from acquisition of pandemic infection of children*. **Nature Communications**, 15, 3210. (2024)
14. Cheng, S. M. S., Mok, C. K. P., Li, J. K. C., Chan, K. K. P., Luk, K. S., Lee, B. H. W., **Gu, H.**, Chan, K. C. K., Tsang, L. C. H., Yiu, K. Y. S., Ling, K. K. C., Tang, Y. S., Luk, L. L. H., Yu, J. K. M., Pekosz, A., Webby, R. J., Cowling, B. J., Hui, D. S. C. & Peiris, M. *Cross-neutralizing antibody against emerging Omicron subvariants of SARS-CoV-2 in infection-naïve individuals with homologous BNT162b2 or BNT162b2 (WT+ BA. 4/5) bivalent booster vaccination*. **Virology Journal**, 21(1), 1–7. (2024)
15. Xu, X., Deng, Y., Ding, J., Shi, X., Zheng, X., Wang, D., Yang, Y., Liu, L., Wang, C., Li, S., **Gu, H.**, Poon, L. L. M. 2023 & Zhang, T. *Refining detection methods for emerging SARS-CoV-2 mutants in wastewater: A case study on the Omicron variants*. **Science of The Total Environment**, 904, 166215. (2023)
16. Xu, X., Deng, Y., Ding, J., Zheng, X., Wang, C., Wang, D., Liu, L., **Gu, H.**, Peiris, M., Poon, L. L. M. & Zhang, T. *Wastewater genomic sequencing for SARS-CoV-2 variants surveillance in wastewater-based epidemiology applications*. **Water Research**, 244, 120444. (2023)

17. Zheng, M. Z. M., Tan, T. K., Villalon-Letelier, F., Lau, H., Deng, Y.-M., Fritzlar, S., Valkenburg, S. A., **Gu, H.**, Poon, L. L. M., Reading, P. C., Townsend, A. R. & Wakim, L. M. *Single-cycle influenza virus vaccine generates lung CD8+ Trm that cross react against viral variants and subvert virus escape mutants*. **Science Advances**, 9 (36), eadg3469. (2023)
18. Xie, R., Edwards, K. M., Adam, D. C., Leung, K. S. M., Tsang, T. K., Gurung, S., Xiong, W., Wei, X., Ng, D. Y. M., Liu, G. Y. Z., Krishnan, P., Chang, L. D. J., Cheng, S. M. S., **Gu, H.**, Siu, G. K. H., Wu, J. T., Leung, G. M., Peiris, M., Cowling, B. J., Poon, L. L. M. & Dhanasekaran, V. *Resurgence of Omicron BA.2 in SARS-CoV-2 infection-naïve Hong Kong*. **Nature Communications**, 14, 2422. (2023)
19. **Gu, H.**, Quadeer, A. A., Krishnan, P., Chang, L., Liu, G., Ng, D., Cheng, S., Lam, T., Peiris, M., McKay, M. & Poon, L. *Within-host genetic diversity of SARS-CoV-2 lineages in unvaccinated and vaccinated individuals*. **Nature Communications**, 14, 1793. (2023)
20. Adam, D., Martín-Sánchez, M., **Gu, H.**, Wu, P., Lau, E., Leung, G., Poon, L. & Cowling, B. *Risk of within-hotel transmission of SARS-CoV-2 during on-arrival quarantine in Hong Kong: an epidemiological and phylogenomic investigation*. **The Lancet Regional Health - Western Pacific**, 100678. (2023)
21. Chu, J. T. S., **Gu, H.**, Sun, W., Fan, R. L. Y., Nicholls, J. M., Valkenburg, S. A. & Poon, L. *Heterosubtypic immune pressure accelerates emergence of influenza A virus escape phenotypes in mice*. **Virus Research**, 198991. (2022)
22. Leong, N., **Gu, H.**, Ng, Y. M., Chang, L., Krishnan, P., Cheng, S., Peiris, M. & Poon, L. *Development of multiplex RT-ddPCR assays for detection of SARS-CoV-2 and other common respiratory virus infections*. **Influenza and Other Respiratory Viruses**, 1 (11), 13084. (2022)
23. Jia, J., Tan, C. W., Cheng, S., **Gu, H.**, Yeoh, A., Mok, C. K. P., Wang, Y., Zhao, J., Leung, N., Cowling, B., Poon, L., Hui, D., Wang, L.-F., Peiris, M. & Valkenburg, S. *Priming conditions shape breadth of neutralizing antibody responses to sarbecoviruses*. **Nature Communications**, 13, 6285. (2022)
24. Hui, K., Ng, K.-C., Ho, J. C. W., Yeung, H.-W., Ching, R. H. H., **Gu, H.**, Chung, J. C. K., Chow, V. L. Y., Sit, K.-Y., Hsin, M. K. Y., Au, T. W. K., Poon, L., Peiris, J. S., Nicholls, J. & Chan, M. *Replication of SARS-CoV-2 Omicron BA. 2 Variant in Ex Vivo Cultures of the Human Upper and Lower Respiratory Tract*. **eBioMedicine**, vol. 83: 104232. (2022)
25. Su, W., Choy, K. T., **Gu, H.**, Sia, S. F., Cheng, K. M., Nizami, S. I. N., Krishnan, P., Ng, Y. M., Chang, L., Liu, Y., Cheng, S., Peiris, M., Poon, L., Nicholls, J. & Yen, H.-L. *Reduced Pathogenicity and Transmission Potential of Omicron BA.1 and BA.2 Sublineages Compared with the Early Severe Acute Respiratory Syndrome Coronavirus 2 D614G Variant in Syrian Hamsters*. **Journal of Infectious Diseases**, jiac276. (2022)
26. Sun, W., Cheng, S., Lam, K., Kwan, T., Wong, R., Lau, L., Liu, G., Luk, L., Li, J., **Gu, H.**, Peiris, M. & Poon, L. *Natural Reassortment of Eurasian Avian-Like Swine H1N1 and Avian H9N2 Influenza Viruses in Pigs, China*. **Emerging Infectious Diseases**, 28.7: 1509. (2022)
27. Bull, M. B., **Gu, H.**, Ma, F., Perera, L., Poon, L. & Valkenburg, S. *Next Generation T cell-activating vaccination increases influenza virus mutation prevalence*. **Science Advances**, 8(14), eabl5209. (2022)
28. Hachim, A., **Gu, H.**, Kaviani, O., Mori, M., Kwan, M., Chan, W. H., Yau, Y. S., Chiu, S., Tsang, O., Hui, D., Mok, C., Ma, F., Lau, E., Amarasinghe, G., Qavi, A., Cheng, S., Poon, L., Peiris, M., Valkenburg, S. & Kaviani, N. *SARS-CoV-2 accessory proteins reveal distinct serological signatures in children*. **Nature Communications**, 13: 2951. (2022)
29. Yen, H.-L., Sit, T., Brackman, C., Chuk, S., **Gu, H.**, Tam, K., Law, P., Leung, G., Peiris, M., Poon, L., Cheng, S., Chang, L., Krishnan, P., Ng, D., Liu, G., Hui, M., Ho, S. Y., Su, W., Sia, S. F., Choy, K.-T., Cheuk, S., Lau, S., Tang, A., Koo, J. & Yung, L. *Transmission of SARS-CoV-2 delta variant (AY. 127) from pet hamsters to humans, leading to onward human-to-human transmission: a case study*. **The Lancet**, 399.10329: 1070–1078. (2022)
30. **Gu, H.\***, Xie, R.\*, Adam, D.\*, Tsui, J.\*, Chu, D.\*, Chang, L., Cheuk, S., Gurung, S., Krishnan, P., Ng, D., Liu, G., Wan, C., Cheng, S., Edwards, K., Leung, K., Wu, J., Tsang, D., Leung, G., Cowling, B., Peiris, M., Lam, T., Dhanasekaran, V. & Poon, L. *Genomic epidemiology of SARS-CoV-2 under an elimination strategy in Hong Kong*. **Nature Communications**, 13, 736 : 1–10. (2022)
31. Hui, K., Ho, J., Cheung, M.-C., Ng, K.-C., Ching, R., Lai, K.-L., Kam, T., **Gu, H.**, Sit, K.-Y., Hsin, M., Au, T., Poon, L., Peiris, M., Nicholls, J. & Chan, M. *SARS-CoV-2 Omicron variant replication in human respiratory tract ex vivo*. **Nature**, 603: 715–720. (2022)
32. **Gu, H.**, Ng, D., Liu, G., Cheng, S., Krishnan, P., Chang, L., Cheuk, S., Hui, M., Lam, T., Peiris, M. & Poon, L. *Recombinant BA. 1/BA. 2 SARS-CoV-2 Virus in Arriving Travelers, Hong Kong, February 2022*. **Emerging infectious diseases**, 28 (6) : 1276. (2022)
33. **Gu, H.**, Krishnan, P., Ng, D., Chang, L., Liu, G., Cheng, S., Hui, M., Fan, M., Wan, J., Lau, L., Cowling, B., Peiris, M. & Poon, L. *Probable Transmission of SARS-CoV-2 Omicron Variant in Quarantine Hotel, Hong Kong, China, November 2021*. **Emerging infectious diseases**, 28 (2): 460. (2022)
34. **Gu, H.**, Cheng, S., Krishnan, P., Ng, D., Chang, L., Liu, G., Cheuk, S., Hui, M., Fan, M., Wan, J., Lau, L., Chu, D., Dhanasekaran, V., Peiris, M. & Poon, L. *Monitoring International Travelers Arriving in Hong Kong for Genomic Surveillance of SARS-CoV-2*. **Emerging infectious diseases**, 28(1): 247–250. (2022)

35. **Gu, H.**, Chu, D., Chang, L., Cheuk, S., Gurung, S., Krishnan, P., Ng, D., Liu, G., Wan, C., Xie, R., Cheng, S., Cowling, B., Tsang, D., Peiris, M., Dhanasekaran, V. & Poon, L. *Genetic Diversity of SARS-CoV-2 among Travelers Arriving in Hong Kong. Emerging infectious diseases*, 27(10): 2666–2668. (2021)
36. Dhanasekaran, V., Edwards, K., Xie, R., **Gu, H.**, Adam, D. C., Chang, L., Cheuk, S., Gurung, S., Krishnan, P., Ng, D., Liu, G., Wan, C., Cheng, S., Tsang, D., Cowling, B., Peiris, M. & Poon, L. *Air travel-related outbreak of multiple SARS-CoV-2 variants. Journal of Travel Medicine*, taab149. (2021)
37. Chu, D.\*, **Gu, H.\***, Chang, L., Cheuk, S., Gurung, S., Krishnan, P., Ng, D., Liu, G., Wan, C., Tsang, D., Peiris, M. & Poon, L. *SARS-CoV-2 superspread in fitness center, Hong Kong, China, March 2021. Emerging infectious diseases*, 27(8): 2230–2232. (2021)
38. Chu, D.\*, Hui, K.\*, **Gu, H.**, Ko, R. L. W., Krishnan, P., Ng, D., Liu, G., Wan, C., Cheung, M.-C., Ng, K.-C., Nicholls, J., Tsang, D., Peiris, M., Chan, M. & Poon, L. *Introduction of ORF3a-Q57H SARS-CoV-2 variant causing fourth epidemic wave of COVID-19, Hong Kong, China. Emerging infectious diseases*, 27(5): 1492–1495. (2021)
39. **Gu, H.**, Chu, D., Peiris, M. & Poon, L. *Multivariate analyses of codon usage of SARS-CoV-2 and other betacoronaviruses. 2020 Virus Evolution* 6, no. 1: veaa032. (2020)
40. Sah, R., Rodriguez-Morales, A. J., Jha, R., Chu, D., **Gu, H.**, Peiris, M., Bastola, A., Lal, B. K., Ojha, H. C., Rabaan, A. A., Zambrano, L. I., Costello, A., Morita, K., Pandey, B. D. & Poon, L. *Complete genome sequence of a 2019 novel coronavirus (SARS-CoV-2) strain isolated in Nepal. Microbiology Resource Announcements* 9, no. 11. (2020)
41. **Gu, H.**, Fan, R., Wang, D. & Poon, L. *Dinucleotide evolutionary dynamics in influenza A virus. Virus Evolution* 5, no. 2: vez038. (2019)
42. Hussain, A.-G., Chu, D., Rihan, E. M. A., Abd-Allah, E. M., **Gu, H.**, Chin, A., Qasim, I. A., Aldoweriej, A., Alharbi, S. 2018 S., Al-Aqil, M. A., Al-Sahaf, A., Abdel Rahman, S. S., Aljasseem, A. H., Abdul-Al, A., Aljasir, M. R., Alhammad, Y. M. O., Kasem, S., Peiris, M., Zaki, A. Z. S. A. & Poon, L. *Circulation of Influenza A (H5N8) Virus, Saudi Arabia. Emerging Infectious Diseases* 24, no. 10: 1961. (2018)
43. **Gu, H.**, Leung, R. K.-K., Jing, Q., Zhang, W., Yang, Z., Lu, J., Hao, Y. & Zhang, D. *Meteorological Factors for Precise Dengue Fever Control and Prevention in South China. International Journal of Environmental Research and Public Health*, 13(9), 867. (2016)
44. Chen, T.\*, **Gu, H.\***, Leung, R. K.-K., Liu, R., Chen, Q., Wu, Y. & Li, Y. *Evidence-Based interventions of Norovirus outbreaks in China. BMC Public Health*, 16(1): 1072. (2016)
45. **Gu, H.**, Zhang, W., Xu, H., Li, P., Wu, L., Guo, P., Hao, Y., Lu, J. & Zhang, D. *Predicating risk area of human infection with avian influenza a (H7N9) virus by using early warning model in China. Chinese Journal of Epidemiology*, 36(5): 470–475. (2015)
46. Li, P., Xu, H., **Gu, H.**, Ma, M., Lu, J., Hao, Y. & Zhang, D. *Epidemiological Characteristics of Influenza Virus H7N9 and Evolutionary Analysis of Hemagglutinin and Neuraminidase Genes. Journal of Sun Yat-sen University (Medical Sciences)*, 06. (2014)

## SUPERVISION

### Co-supervision:

- |                                                                                                                                                                                                                                                                    |                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Ruixuan Wang, PhD student, The University of Hong Kong<br/><i>Within-host Viral Phylodynamics</i></li> <li>• Mengting Li, PhD student, The University of Hong Kong<br/><i>Virtual Cell and Viral Infection</i></li> </ul> | <p>Feb. 2025 – May 2026</p> <p>Oct. 2025 – May 2026</p> |
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## INVITED TALKS

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| <ul style="list-style-type: none"> <li>• <b>Genomic Surveillance and Molecular Epidemiology of SARS-CoV-2 in Hong Kong.</b><br/><i>XLII Annual Meeting of the Willi Hennig Society, The Chinese University of Hong Kong, HKSAR, China</i></li> </ul> | <p>Jul. 2025</p> |
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## PROFESSIONAL DEVELOPMENT

### Workshops:

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| <ul style="list-style-type: none"> <li>• <i>Workshop on Molecular Evolution</i>, Marine Biological Laboratory (MBL), Woods Hole, MA, USA</li> <li>• <i>Virus Evolution and Epidemiology Summer Course</i>, Croucher Foundation, Hong Kong</li> <li>• <i>Models and Inference in Population Genetics IV</i>, University of Warwick, UK</li> </ul> | <p>May 2025</p> <p>Jul. 2024</p> <p>Apr. 2024</p> |
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### Online specializations:

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| <ul style="list-style-type: none"> <li>• <i>Discrete Mathematics for Computer Science</i>, University of California San Diego</li> <li>• <i>Bayesian Statistics and Modelling</i>, University of California Santa Cruz</li> <li>• <i>Python 3 Programming</i>, University of Michigan</li> </ul> | <p>Oct. 2019</p> <p>Aug. 2019</p> <p>Jun. 2019</p> |
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- *Mathematics for Machine Learning*, Imperial College London

Apr. 2019

#### Teaching aptitude:

- Certificate of Teaching and Learning in Higher Education, CETL, HKU
- PBL Tutor Training Certificate, LKS Faculty of Medicine, The University of Hong Kong

Dec. 2024

Oct. 2025

### ACADEMIC SERVICES

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#### Managing editor:

**Virology Journal**

#### Ad hoc peer reviewer:

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|-----------------------------------------|-----------------------------------------|
| SCIENCE                                 | THE LANCET MICROBE                      |
| COMMUNICATIONS MEDICINE                 | EMERGING INFECTIOUS DISEASES            |
| PLOS PATHOGENS                          | CELLULAR AND MOLECULAR LIFE SCIENCES    |
| FRONTIERS IN IMMUNOLOGY                 | INFLUENZA AND OTHER RESPIRATORY VIRUSES |
| VIRUS EVOLUTION                         | SCIENTIFIC REPORTS                      |
| NPJ VIRUSES                             | EPIDEMICS                               |
| VIROLOGY JOURNAL                        | BMC MICROBIOLOGY                        |
| BMC MEDICAL RESEARCH METHODOLOGY        | INFECTION, GENETICS AND EVOLUTION       |
| JOURNAL OF MULTIDISCIPLINARY HEALTHCARE |                                         |

#### Open-source software:

- **Author**, *SynMut*, *Synonymously Mutated Sequences with Different Genomic Signatures*
- **Author**, *GEII*, *Host Gene Expression under Influenza virus infection or Interferon stimulation*
- **Author**, *epitope\_diversity*, *Estimating epitope diversity at haplotype level using NGS reads*
- **Author**, *fasta\_filter*, *Filtering fasta sequences with threshold of specific bases*
- **Contributor**, *spatPomp*, *Inference for Spatiotemporal Partially Observed Markov Processes*
- **Contributor**, *data.table* (Chinese translation), *High-performance R package*

#### Professional leadership:

- Founder and sole lecturer of a weekly reading group on Molecular Evolution. Feb. 2025 – Dec. 2025  
*Personally delivered every session, teaching Ziheng Yang's "Molecular Evolution: The Statistical Approach" to 20 researchers from different regions and disciplines*
- Generating and drafting weekly reports to Hong Kong public health authorities. 2020 – 2022  
*On SARS-CoV-2 genomic epidemiology for timely pandemic responses*